COVID-19 Case Study

(Chicago, Illinois)

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Introduction

COVID-19 revealed strong **geographic and social disparities** across Chicago communitiesStructural factors (race/ethnicity, poverty) may influence **community-level transmission**Mass incarceration is an important **contextual driver of health inequities**, even if not directly measured hereNeed to understand how **community characteristics relate to COVID-19 case rates** at the ZIP code level (see references)

Data and Methods

Data Source: Data obtained from the Harvard Dataverse, inspired by a study using Cook County Jail and community-level COVID-19 dataAnalysis uses ZIP code–level COVID-19 cumulative case rates in ChicagoIncludes demographic and socioeconomic variables (race/ethnicity, poverty, population density, public transit use)COVID-19 data reflect community-level transmission, not individual or jail-level outcomesJail data informed the study motivation, but were not directly analyzed in this project

Key variables:

1. **Cumulative cases per capita**
2. **Black_pct (percent population)**
3. **Hisp_pct**
4. **(Population) Density**
5. **Poverty**
6. **Public transit use**

Analysis methods: Initial analysis was conducted using multiple linear regression. Final analysis was completed using the poisson generalized linear model (GLM)

Results

Table 1. Descriptive Statistics by ZIP Code

	Missing	Overall
n		58
ZIP Code, mean (SD)	0	60634.6 (32.7)
Total Population, mean (SD)	0	47824.4 (26101.6)
Percent Black, mean (SD)	0	0.3 (0.3)
Percent Hispanic, mean (SD)	0	0.2 (0.2)
Percent White, mean (SD)	0	0.5 (0.3)
Poverty Rate, mean (SD)	0	0.2 (0.2)
Population Density, mean (SD)	0	16001.4 (9375.8)
Public Transit Use, mean (SD)	0	0.3 (0.1)
Weekly Case Rate, mean (SD)	0	88.6 (40.9)
Weekly Death Rate, mean (SD)	0	3.8 (2.6)
Weekly Test Rate, mean (SD)	0	774.1 (214.8)
Percent Positive, mean (SD)	0	0.1 (0.1)
Cumulative Case Rate, mean (SD)	0	2064.9 (936.6)

Table 2. Multiple Linear Regression

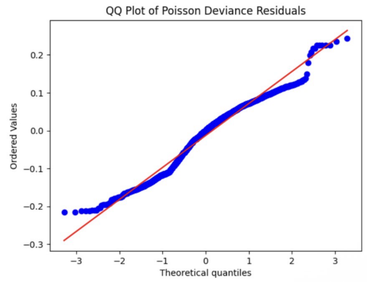
	coef	std err	t	P> |t|	[0.025	0.975]
Intercept	0.0049	0.001	5.516	0.000	0.003	0.007
black_pct	0.0031	0.002	1.949	0.052	-2.04e-05	0.006
hisp_pct	0.0158	0.002	9.582	0.000	0.013	0.019
density	-3.184e-09	3.53e-08	-0.090	0.928	-7.25e-08	6.61e-08
poverty	0.0105	0.003	3.257	0.001	0.004	0.017
public_transit	-0.0017	0.003	-0.573	0.566	-0.008	0.004

Omnibus:	7.437	Durbin-Watson:	0.717
Prob(Omnibus):	0.024	Jarque-Bera (JB):	7.517
Skew:	0.183	Prob(JB):	0.0233
Kurtosis:	2.963	Cond. No.	3.03e+05

Table 3. Poisson Linear Model

	coef	std err	z	P> |z|	[0.025	0.975]
Intercept	-5.0699	1.058	-4.792	0.000	-7.144	-2.996
black_pct	0.2529	1.725	0.147	0.883	-3.129	3.635
hisp_pct	1.2176	1.665	0.731	0.465	-2.046	4.481
density	-2.929e-06	4.33e-05	-0.068	0.946	-8.79e-05	8.2e-05
poverty	1.0218	3.480	0.294	0.769	-5.799	7.843
public_transit	-0.0727	3.699	-0.020	0.984	-7.322	7.176

QQ Plot of Poisson Deviance Residuals



Results

- OLS residuals violate normality (QQ plot deviation)
- Skewness indicates poor model fitSome predictors significant in OLS, but assumptions not met
- Poisson model better suited for outcome type, but yields no significant effects
- Conclusions depend strongly on model selection
- Small sample size (n = 58) may limit statistical power and stability of estimates
- Potential multicollinearity or unmeasured confounding may affect coefficient reliability

Conclusions/Future Directions

Higher **Hispanic population percentage** is positively associated with higher case rates. Higher **poverty rates** are also positively associated with higher case rates

Public health impact

Highlights the importance of using appropriate models for disease data to avoid misleading conclusions that could affect resource allocation and policy decisions

Future direction & limitation

Use larger datasets and consider alternative models (e.g., negative binomial, spatial models) to better capture variability and geographic effects

Limitation

Small sample size and potential unmeasured confounding limit the reliability and generalizability of results

References

Reinhart E, Chen DL. Carceral-community epidemiology, structural racism, and COVID-19 disparities. Proc Natl Acad Sci U S A. 2021 May 25;118(21):e2026577118. doi: 10.1073/pnas.2026577118. PMID: 33972409; PMCID: PMC8166074.

Reinhart E, Chen DL. Carceral-community epidemiology, structural racism, and COVID-19 disparities. Proc Natl Acad Sci U S A. 2021 May 25;118(21):e2026577118. doi: 10.1073/pnas.2026577118. PMID: 33972409; PMCID: PMC8166074.